# Muhammad Rommel Abbas

**ML / Computer Vision / Software Engineering**

Rochester, NY | Summer 2027 internship candidate

**PORTFOLIO WEBSITE**
**rommelabbas.github.io**
View projects, demos, and work


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## EDUCATION

**Rochester Institute of Technology** — Expected May 2030
**B.S. in Computer Science** — Rochester, NY

## TECHNICAL SKILLS

**Languages:** Python, Java, SQL, Bash
**ML / Vision:** PyTorch, scikit-learn, XGBoost, pandas, Ultralytics YOLO, OpenCV, VLMs, face recognition
**LLMs / Speech:** RAG, GraphRAG, LangChain, FAISS, Ollama, llama.cpp, CUDA, faster-whisper, Coqui XTTS, speaker ID
**Backend / Systems:** Django, FastAPI, PostgreSQL, SQLite, Neo4j, Docker, Linux, Nginx, Git, GitHub Actions, pytest
**Robotics / Embedded:** ROS 2, Nav2, Raspberry Pi, ESP32, Intel RealSense, motor control

## RESEARCH

**Undergraduate Researcher | Prof. John Thomas's Group, RIT** — Oct 2026 - Present

- Sole member on Track 1 (EEG-to-Image) of the **NeurIPS 2026 EEG/EMG Foundation Challenge**, working on image decoding from brain signals.
- Running experiments on a local **RTX 5080** using the lab EEG preprocessing pipeline, with access to a **96 GB GPU server** for larger models.

## EXPERIENCE

**Machine Learning / Computer Vision Intern** — Jun-Aug 2025
**Punjab Safe Cities Authority | Government of Punjab** — Lahore, Pakistan

- Trained YOLO-based models to detect urban safety hazards, including open manholes; annotated and preprocessed camera data for an authority operating **12,000+ cameras**.
- Built an **event-triggered encoder-to-VLM pipeline** for camera feeds, invoking vision-language inference only when events were detected to avoid redundant model calls.
- Implemented a multi-stage **face-recognition and identity-matching** pipeline for camera imagery, linking visual detection with downstream identity analysis.

**Cybersecurity Intern** — Aug-Sep 2025
**Counter Terrorism Department (CTD) | Government of KP** — Peshawar, Pakistan

- Conducted **open-source intelligence (OSINT)** analysis of extremist activity to inform threat assessments for public gatherings, and presented synthesized findings to senior analysts.
- Received a **Class II Certificate of Commendation** from the Additional Inspector General of Police, Khyber Pakhtunkhwa.

## PROJECTS

**Regional Car Price Prediction** — GitHub ↗

- Built an end-to-end valuation app with a resumable scraper for **59,668 PakWheels listings**, curating a deduplicated **48K-row dataset** for regional used-car modeling.
- Trained XGBoost with a leakage-aware split: **0.835 $R^2$, 11.54% MAPE** on a **9,674-row holdout**; **86.1%** of predictions within 20% of asking price.
- Deployed via Django with **20 passing tests, 81% coverage**, and GitHub Actions CI across Python 3.11-3.13.

**ALIRA - Local Voice AI Assistant & Robot** — Dec 2024 - Present

- Developed an installable PyTorch wake-word prototype, moving from a 239,745-parameter LSTM to a **39,969-parameter CNN** (~83% fewer parameters) with 80-bin Mel features.
- Built an offline speech stack using **Silero VAD** and **DeepFilterNet3**; benchmarked emotion2vec at a **0.05 real-time factor** on CPU to evaluate local audio inference.
- Developing **Neo4j GraphRAG** that combines entity traversal and vector retrieval; integrating face and speaker identification to connect users with conversation history.
- Building a **ROS 2 robot** with Raspberry Pi audio, GPU inference, encoder odometry, and planned Nav2/RealSense navigation.

**SkillTree - Tech Job Market Intelligence** — skilltree.handoverit.com

- Built a Django analytics platform covering **69,491 job postings from 44+ sources**; normalized technical skills using Llama models via Groq for searchable labor-market insights.
- Stored results in SQLite and self-hosted the service on Linux using **Gunicorn, Nginx, and Cloudflare Tunnel**.

**More GitHub work: Local LLM Inference** - tuned llama.cpp/CUDA for Qwen3-Coder 30B on RTX 5080 with MoE CPU expert offloading.
**Hermes:** self-hosted Discord/Matrix agent with ESP32 display; **WhisperKey:** private offline Linux dictation using faster-whisper.

## LEADERSHIP & INVOLVEMENT

**Computer Science House (CSH), RIT** — Aug 2026 - Present

- Selected as an off-floor member; built a read-only **Slack-to-local-retrieval bot** indexing channel history for contextual search.
- Invited to deliver an R&D seminar on **local LLM inference** and voice-assistant architecture.

**RITSEC (RIT Cybersecurity Club)** — Aug 2026 - Present

- Active member; attend weekly technical sessions on **offensive security and CTF tooling**.